Polymer Chemistry

PAPER

View Article Online
View Journal | View Issue

Cite this: *Polym. Chem.*, 2016, **7**, 7506

Separation and characterization of benzaldehyde-functional polyethylene glycols by liquid chromatography under critical conditions†

Y.-Z. Wei,[a] Y.-F. Chu,[a] E. Uliyanchenko,‡[b] P. J. Schoenmakers,[b] R.-X. Zhuo[a] and X.-L. Jiang*[a,b]

Received 20th September 2016, Accepted 16th November 2016

DOI: 10.1039/c6py01653g

www.rsc.org/polymers

The distributions of non-, mono- and bi-functional benzaldehyde-substituted polyethylene glycols (PEGs) need to be accurately determined for biomedical applications. The baseline separation of PEGs and their derivatives based on the number of functional end groups without an obvious effect of molar mass is achieved by high-performance liquid chromatography (HPLC) and ultra-performance liquid chromatography (UPLC) using reversed-phase packing columns under critical isocratic elution conditions. The effects of organic solvent percentage, column temperature and buffer concentration on the retention times of PEGs and benzaldehyde-substituted PEGs have been investigated. The separation was mainly due to the hydrophobic interaction between the benzaldehyde end-groups and column packing. HPLC analysis can provide useful information to optimize the synthesis of functional PEGs. To our knowledge, this is the first report on separation of benzaldehyde-functional PEGs based on the functionality independent of the molar mass by isocratic elution.

Introduction

Polyethylene glycol (PEG) and its derivatives have been widely applied in drug delivery systems and in the biopharmaceutical industry, forming conjugates with synthetic polymers, peptides, proteins, lipids, oligonucleotides, and small molecule drugs.[1–6] Based on the pioneering work of Abuchowski *et al.*[7] in the late 1970s, the covalent conjugation of PEG (PEGylation) has emerged as a valuable tool to overcome many of the deficiencies, particularly of protein- and peptide-based drugs, by increasing bioavailability (circulation time) and shielding them from proteolytic degradation and immune response.[8,9] Bioconjugation of PEG derivatives is often achieved *via* reactive functional end groups, such as carboxyl, amine, thiol, azide, vinyl sulfone, acetylene, acrylate and aldehyde groups.[10,11] For instance, an aldehyde group can react with an amino group to form pH sensitive imine bonds that remain stable at physiological pH, but are cleavable at acidic pH.[12,13] Aldehyde-terminated PEGs are widely studied in tissue engineering and drug delivery systems.[14–16] Chitosan-based micelles were modified with mono-aldehyde-functionalized PEGs for anti-cancer drug delivery, which significantly reduced the macrophage uptake of the micelles, but did not affect cellular uptake by liver tumor cells.[17] In this case, the non-functionalized PEGs would not play the modification role. Bi-aldehyde-functionalized PEGs can be used as a cross-linker.[18] Wei *et al.*[19] had studied chitosan-based hydrogels for controlled release of bioactive molecules. They pointed out that only bi-benzaldehyde substituted PEGs can react with chitosan to form gel, and no gel formation was observed when non-functionalized PEGs or mono-benzaldehyde substituted PEGs were used as controls. In this case, the unreacted PEGs and mono-functionalized PEGs are undesired impurities, which are very difficult to remove. However, it is very difficult to produce these and other telechelic well-defined polymers with the desired end-groups. Therefore, the accurate determination of the distribution of non-, mono-, and bi-functionalized PEGs is very crucial for biomedical applications.

Reversed-phase liquid chromatography (RPLC) is by far the most widely used mode of high-performance liquid chromatography (HPLC). Recently, liquid chromatography under critical conditions (LCCC) has been proven to be effective in the analysis of functional polymers.[20–23] Under the critical conditions, the polymers elute at the same time regardless of the molar

[a]*Key Laboratory of Biomedical Polymers of Ministry of Education & Department of Chemistry, Wuhan University, Wuhan 430072, P. R. China. E-mail: xljiang@whu.edu.cn*

[b]*Analytical-Chemistry Group, Faculty of Science, van't Hoff Institute for Molecular Sciences, University of Amsterdam, Science Park 904, 1098 XH Amsterdam, The Netherlands. E-mail: e.uliyanchenko@uva.nl, p.j.schoenmakers@uva.nl*

†Electronic supplementary information (ESI) available. See DOI: 10.1039/c6py01653g

‡Current address: Analytical Technology, Corporate T&I, SABIC, Plasticslaan 1, 4612 PX Bergen op Zoom, The Netherlands. E-mail: elena.uliyanchenko@sabic.com

mass. The separation then only depends on the functional end groups so that the end-group separation of various polymers can be realized.[24–26] A large number of papers have been published demonstrating the possibilities of critical chromatography to separate polymers based on their end group functionality, independent of the number of repeating backbone units.[27–30] However, as far as we know no paper on separation of unreacted PEGs and benzaldehyde-functionalized PEGs based on functionality has been published.

In this work, we describe a chromatographic method for separation and characterization of PEGs and benzaldehyde-substituted PEGs on reversed-phase packing columns. Factors such as column temperature, organic solvent percentage, and salt concentration in the mobile phase influencing the retention times of PEGs and benzaldehyde-substituted PEGs have been investigated. Normal HPLC and ultra-performance liquid chromatography (UPLC) under the near critical conditions were carried out. The off-line MALDI-TOF mass spectrometry was used to confirm the polymer structure.

Experimental

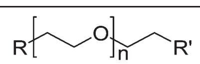
Materials

All the abbreviations of the studied PEGs in this work are explained in Table 1. MPEGOH 5k was purchased from Sigma-Aldrich (LOT no. 81323, M_p 5420, M_n 4677, PDI 1.13, Steinheim, Germany). PEG 2k (average M_n 2000), PEG 4k (average M_n 4000), PEG 6k (average M_n 6000), dicyclohexyl carbodiimide (DCC), 4-dimethylamino pyridine (DMAP), ammonium acetate, dichloromethane (DCM) and diethyl ether were all purchased from Sinopharm Chemical Reagent (Shanghai, China). Isopropanol was purchased from Fuyu Fine Chemical (Tianjin, China). 4-Carboxybenzaldehyde was purchased from Aladdin (Shanghai, China). Acetonitrile was purchased from Tedia Company (HPLC grade, Fairfleld, USA). Water from an Arium® pro UF purification system (Sartorius, Goettingen, Germany) was used. All chemicals and reagents were used as received.

Synthesis of PEG derivatives with benzaldehyde terminal

4-Carboxybenzaldehyde (0.18 g, 1.25 mmol), DCC (0.31 g, 1.5 mmol), and DMAP (0.15 g, 1.25 mmol) were added to a solution of PEG 6k (1.5 g, 0.25 mmol) in DCM (30 mL). After being stirred at 30 °C for 24 h, the solution was filtered. The filtrate was concentrated before dissolving in isopropanol (30 mL), and then cooled at 4 °C for 3 h. The resulting crystals (bi-functional PEG 6k) were collected by filtration and washed with isopropanol and diethyl ether with 79% yield. Other PEGs with different molar masses and functional degrees were synthesized according to the same process, except that the feed ratio of 4-carboxybenzaldehyde to PEG was different. As an example for synthesis of different functionalities of MPEGCHO 5k, the feed ratios of 4-carboxybenzaldehyde [COOH] to MPEGOH 5k [OH] and the catalysts used are shown in Table S1.†

Table 1 Structures and abbreviations of the studied PEGs



Compound	R	R′	Abbreviations
1	OH	OH	PEG 2k
2	OCO-Ph-CHO	OCO-Ph-CHO	PEGDCHO 2k
3	OH	OH	PEG 4k
4	OH	OCO-Ph-CHO	PEGCHO 4k
5	OCO-Ph-CHO	OCO-Ph-CHO	PEGDCHO 4k
6	CH_3O	OH	MPEGOH 5k
7	CH_3O	OCO-Ph-CHO	MPEGCHO 5k
8	OH	OH	PEG 6k
9	OH	OCO-Ph-CHO	PEGCHO 6k
10	OCO-Ph-CHO	OCO-Ph-CHO	PEGDCHO 6k

HPLC analysis of PEGs

Analytical chromatography was performed by using an HPLC system assembled with a Cometro 6000 LDI pump (South Plainfield, USA), a temperature controller, an LC-100 UV detector from Wufeng Instruments (Shanghai, China) coupled to a Softa (Model 300 S, Burbank, USA) evaporative light scattering detector (ELSD) on an XB-C18 column (250 mm × 4.6 mm I.D., 5 μm particle size, 300 Å pore size, Welch Materials, Shanghai, China) and an XB-phenyl column (250 × 4.6 mm I.D., 5 μm particle size, 300 Å pore size, Welch Materials, Shanghai, China). Non-functionalized PEGs do not possess any structural elements absorbed in the commonly applied UV region. An ELSD is especially well-suited for the determination of any nonvolatile analyte independent of chemical characteristics.[31] For the ELSD, air was used as a carrier gas, the temperatures of a nebulizer and the drift-tube were set at 50 °C and 80 °C, respectively. The use of a volatile buffer salt was obligatory because of the ELSD, so ammonium acetate was chosen as the buffer salt. Data were collected using EZChrom Elite software. Data analysis was performed on a Microsoft EXCEL spreadsheet.

UPLC analysis was performed on a Waters Acquity UPLC® system (Waters, Milford, MA, USA) on a UPLC BEH C18 Column (100 × 2.1 mm, 1.7 μm particle size, 130 Å pore size, Waters). An Acquity UPLC ELSD Detector (Waters) was used at a nebulizer temperature of 35 °C and a drift-tube temperature of 70 °C. The nebulizing gas was nitrogen at a pressure of 400 kPa. Data were collected and processed with Empower 2 software (Waters).

The mobile phases were prepared by first dissolving appropriate amounts of ammonium acetate in water for the desired concentration. The aqueous phase was filtered through a 0.22 μm HPLC filter before mixing with the organic phase and degassed prior to use by ultrasound. The samples were prepared in acetonitrile : water = 35 : 65 (volume ratio) at 0.2 mg mL^{-1} and injected by a loop volume of 20 μL (3 μL for UPLC). The flow-rate was set at 0.5 mL min^{-1} (1.2 mL min^{-1} for UPLC).

Characterization of PEG derivatives

^{1}H NMR spectra were recorded on a Mercury VX-300 spectrometer (Varian, Palo Alto, USA) at 300 MHz using $CDCl_3$ as a solvent and TMS as an internal standard. MALDI-TOF-MS spectra were collected with an Axima TOF2 mass spectrometer (Shimadzu, Kyoto, Japan). The instrument was equipped with a 337 nm nitrogen laser at a 3 ns pulse width. All the mass spectra were performed in positive ion reflection mode with an accelerating voltage of 20 kV. The polymer fraction obtained from HPLC was mixed with the matrix α-cyano-4-hydroxycinnamic acid (about 1 mg mL^{-1} in acetonitrile/water = 50/50, v/v). Typically, MS spectra were collected by an average of 200 laser shots.

Results and discussion

Synthesis of the benzaldehyde-substituted PEG

The relevant reactions for the synthesis of benzaldehyde-substituted PEGs according to the literature[32] are exemplified in Fig. 1. The end-group functional degrees of PEGs would vary with the feed ratio. The structures of the starting polymer PEGs and the obtained functional PEGs that were used in this study are summarized in Table 1. Fig. 2 shows the representative ^{1}H NMR spectra for functional benzaldehyde-substituted PEGs, both of which have a new peak (b′) at 4.5 ppm, assigned to methylene protons next to the ester bond. This clearly indicates that the benzaldehyde-substituted PEGs were obtained. Although the average functional degree of PEGs can be calculated from the ^{1}H NMR spectrum based on the integral ratio of the peak at 4.5 ppm to the peak at 3.0–4.0 ppm, this method cannot give the specific number of benzaldehyde end-groups of functional PEGs, because the product might contain unreacted PEGs, mono-functional PEGs and bi-functional PEGs. Therefore the HPLC technique has to be used to separate PEGs and functional PEGs based on the end-group and to characterize functional PEGs accurately.

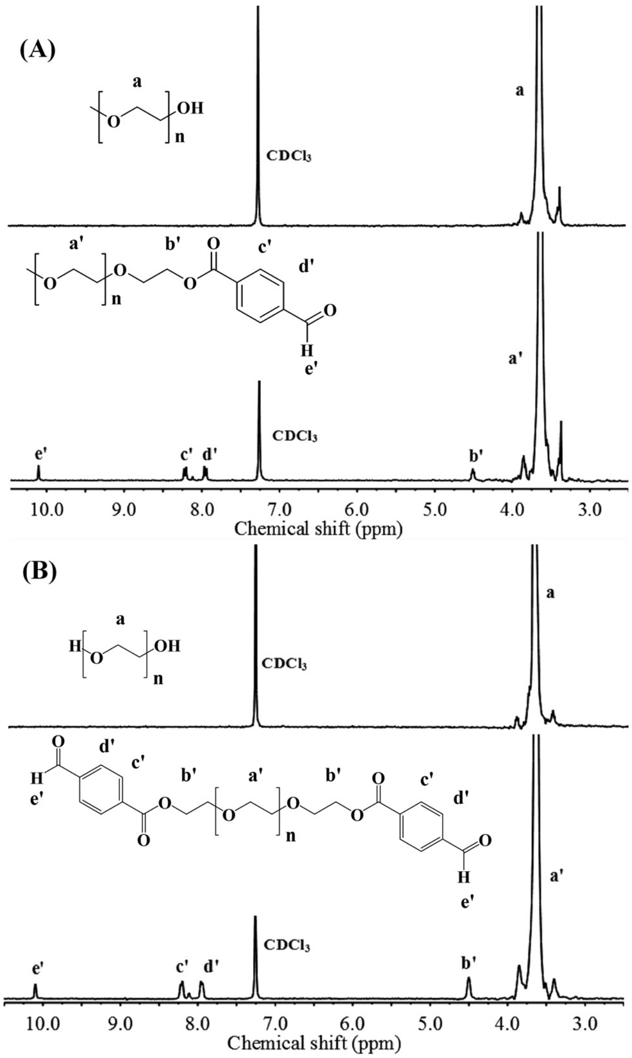


Fig. 2 ^{1}H NMR spectra of benzaldehyde-substituted PEGs and of the starting PEGs in $CDCl_3$. (A) MPEGOH 5k and MPEGCHO 5k and (B) PEG 6k and PEGDCHO 6k.

Effect of organic solvent on the retention of PEG derivatives

Liquid chromatography under critical conditions (LCCC) have been proved to be effective in the analysis of functional polymers.[33] However, the polymer elution behavior in HPLC is very sensitive to column type, mobile phase composition, and column temperature.[34]

The critical compositions for non-functional PEGs were obtained at 40% acetonitrile in water from an XB-C18 column at 25 °C.[35] The effect of the composition of the mobile phase (percentage of acetonitrile in water) on the retention of PEGs and benzaldehyde-substituted PEGs due to the interaction between PEGs and XB-C18 column packing was investigated and is shown in Fig. 3. The retention times for all PEGs and functional PEGs investigated decreased when the mobile phase composition changed from 35 to 55 percent acetonitrile in water. It is well known that elution of PEGs from a C18 column is not possible if no organic additive in water is used as the mobile phase. This indicates that the PEG backbone has weak hydrophobic interactions with the C18 column packing. The addition of acetonitrile decreases the surface tension of water which weakens the hydrophobic interaction between the PEGs and the XB-C18 column packing. It is worth noting that the non-functional PEG 6k and MPEGOH 5k had similar changes in retention times and the bi-functional PEGDCHO 6k had more dramatic changes in retention times than mono-

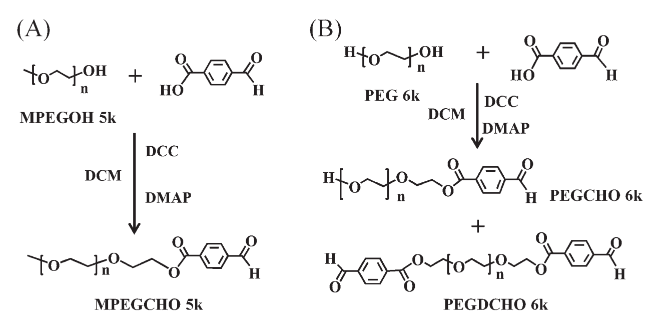


Fig. 1 Synthetic schemes for functional PEGs.

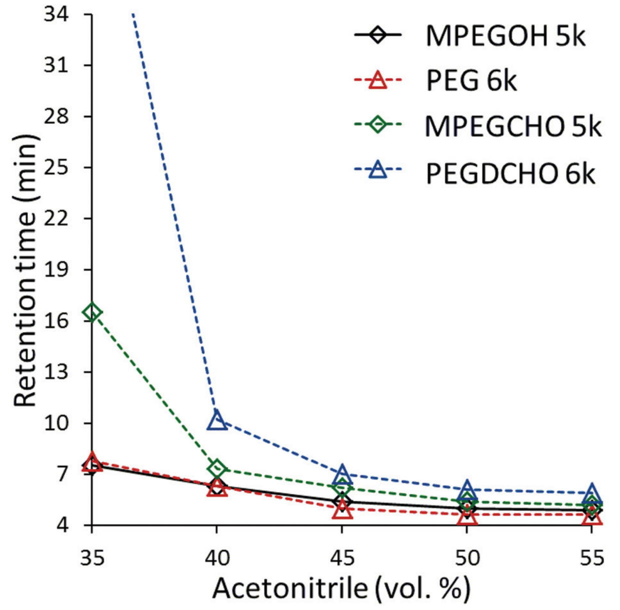


Fig. 3 Effect of acetonitrile volume percentage in mobile phase on retention times of PEGs and benzaldehyde-substituted PEGs on XB-C18 column at 25 °C.

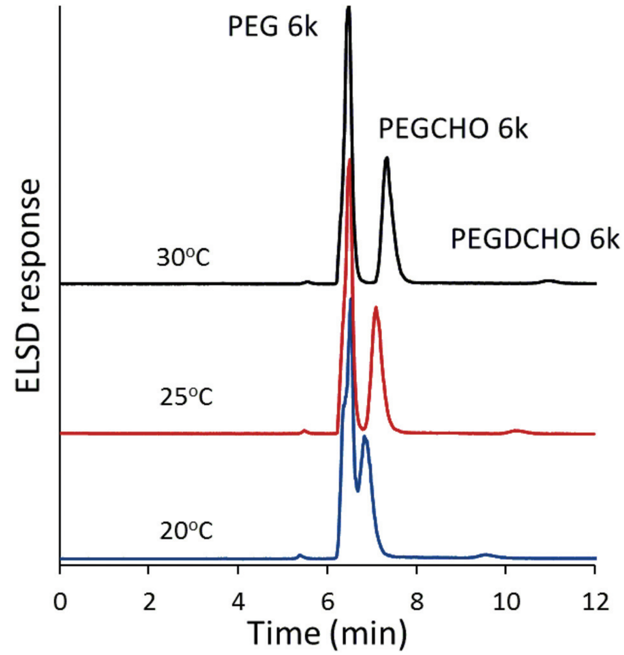


Fig. 4 Effect of column temperature on retention time of PEGs on XB-C18 column (mobile phase: 40% acetonitrile in water).

functional MPEGCHO 5k, when the organic solvent percentage of acetonitrile varied in the mobile phases. This implies that benzaldehyde end-group has stronger hydrophobic interactions with the C18 column packing than the PEG unit. The interaction of the end-group with the stationary phase provides us a good chance to separate PEGs and benzaldehyde-substituted PEGs without the molar mass effect by using the LCCC with the mobile phase 40 vol% acetonitrile in water on the XB-C18 column at 25 °C, which is similar to our previous observation on the Shodex C18 column.[36]

Effects of column temperature on the retention of PEG derivatives

It has been reported that the retention of PEGs increased on increasing the column temperature on the typical reversed phase column in the acetonitrile–water mobile phase.[37–40] Barman *et al.*[41] have found that temperature was a major factor for the separation of PEGs from MPEGOH, and a better resolution between PEGs and MPEGOH could be achieved at a higher column temperature. Therefore, the effect of the column temperature on the retention times of PEGs and benzaldehyde-substituted PEGs was investigated on the XB-C18 column. Fig. 4 shows that a better resolution between PEGs and benzaldehyde-substituted PEGs could be achieved at a higher column temperature in the investigated column temperature range from 20 to 30 °C. The LCCC we obtained above was at 25 °C. It can be seen from Fig. 4 that the baseline separation of the mixture of PEGs and benzaldehyde-substituted PEGs could be achieved when the temperature was set at 25 °C. Thus the temperature of 25 °C was chosen below to study the separation of PEGs and benzaldehyde-substituted PEGs on the XB-C18 column.

Effects of buffer concentration on the retention of PEG derivatives

The hydrophobic interaction could also be enhanced by adding salt to the mobile phase.[42] The effects of different ammonium acetate concentrations on separation of PEGs and benzaldehyde-substituted PEGs were studied on the XB-C18 column, containing 40% acetonitrile in water at 25 °C. As shown in Fig. 5, the retention times for non-functional MPEGOH 5k and PEG 6k were almost the same when the mobile phases containing ammonium acetate in the concentration range from 0 to 0.2 mol L^{-1} were used. However, the retention times of mono-functional MPEGCHO 5k and bi-functional PEGDCHO 6k visibly increased on increasing the ammonium acetate concentration. PEGDCHO 6k had more dramatic changes in retention times than MPEGCHO 5k, when the ammonium acetate concentrations varied in the mobile phases. The reason is that the addition of salt in the water-based mobile phase will increase the hydrophobic interaction. The hydrophobic interaction between the benzaldehyde end-groups and the column packing can be largely enhanced by adding ammonium acetate in the mobile phase. This also provides us an alternative way to separate PEGs and benzaldehyde-substituted PEGs on a typical C18 column without obvious molar mass effects.

Separation of PEG derivatives using LCCC

Successful baseline separation of PEGs and benzaldehyde-substituted PEGs according to the number of benzaldehyde

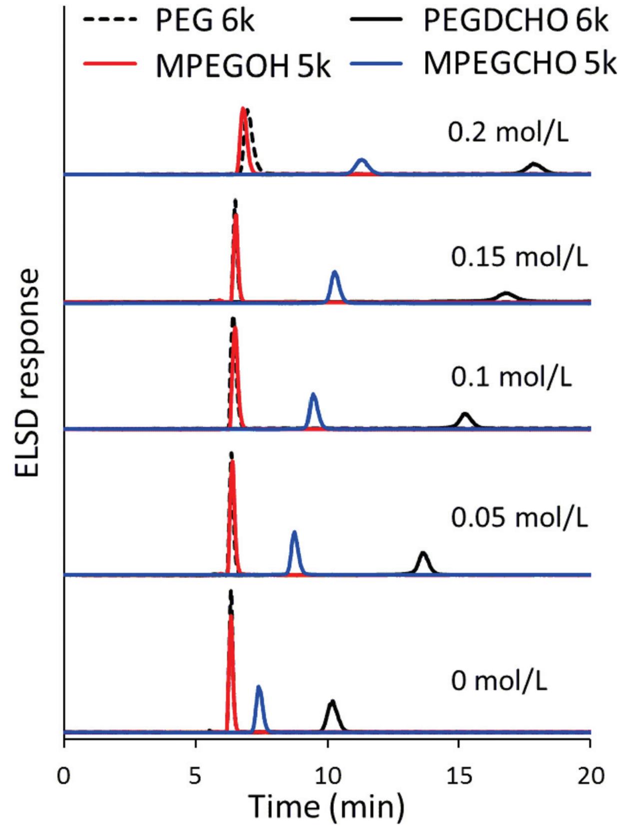


Fig. 5 The chromatograms of PEGs obtained with different ammonium acetate concentrations in the mobile phase (40% acetonitrile in water) on XB-C18 column at 25 °C.

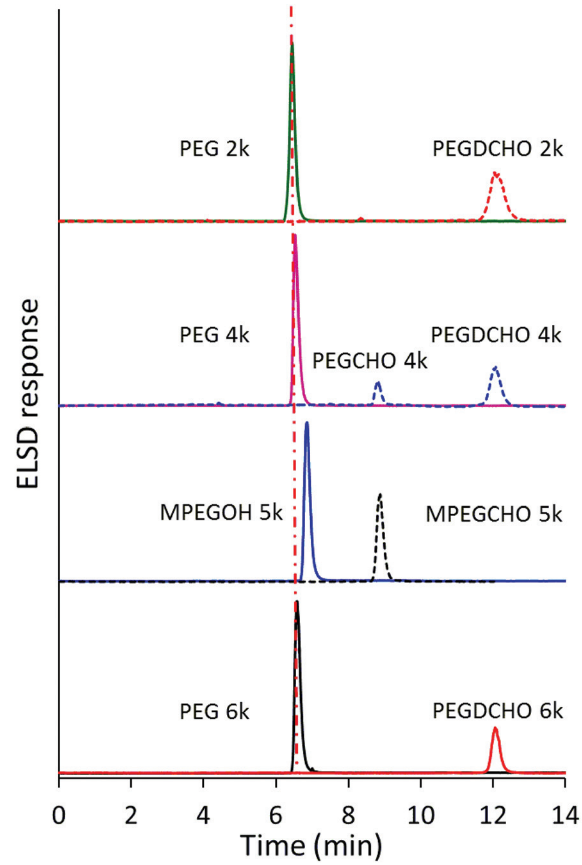


Fig. 6 The chromatograms of PEGs obtained on XB-phenyl column at 30 °C (mobile phase: acetonitrile : water = 45 : 55, v/v).

end-groups was obtained under the LCCC on an XB-C18 column with 40% acetonitrile in water at 25 °C, as shown in Fig. 5 (no ammonium acetate added). PEGDCHO 6k had a longer retention time than MPEGCHO 5k, because the hydrophobic interaction between benzaldehyde the functional end-groups and the column packing became stronger when the number of benzaldehyde end-groups increased. The LCCC for PEGs on the XB-phenyl column was presented at 45% acetonitrile in water at 30 °C.[36] The chromatograms of PEGs and PEG derivatives under this LCCC are exemplified in Fig. 6. It is clearly shown that PEGs and PEG derivatives were successfully separated according to the number of benzaldehyde functional end-groups. Their molar mass had no obvious impact on retention times. For example, the retention times were almost the same for PEG 2k, PEG 4k and PEG 6k. MPEGOH 5k had the longest retention time among the non-benzaldehyde-substituted PEGs investigated, probably due to the hydrophobic effect of the methyl group.[40] The retention times for difunctional PEGDCHO 2k, PEGDCHO 4k and PEGDCHO 6k were also similar. Mono-benzaldehyde-substituted PEGCHO 4k and MPEGCHO 5k had a longer retention time than that of the non-benzaldehyde-substituted PEGs but a shorter retention time than that of the bi-benzaldehyde-substituted PEGs, because the benzaldehyde functional end-group could interact with the phenyl moieties on the column packing through hydrophobic and possible π–π interactions.[43] The interaction became stronger when the number of benzaldehyde-groups increased. The recovery for mono-functional MPEGCHO 5k and bi-functional PEGDCHO 6k was checked according to the method used in reference.[29] The isocratic mobile phase (LCCC, 40 vol% acetonitrile in water) was first employed, then the elution strength of the mobile phase was increased to reach genuine exclusion conditions (60 vol% acetonitrile in water). We only observed peaks before the start of the gradient, which indicated that there was no sample recovery problem in this study. This chromatographic LCCC condition worked well for separation of PEGs based on the number of benzaldehyde functional groups independent of the molar mass on the XB-phenyl column.

To further confirm that this separation is based on the number of the end functional groups of benzaldehyde, it is relatively straightforward to collect fractions and to perform off-line mass spectrometry analysis to identify the repeating units and, especially, the end-groups. As a representative example, the matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS) spectrum of the fraction collected at 8.45–8.70 min (Fig. 6 for sample MPEGCHO 5k) is shown in Fig. 7. Two different series of peaks

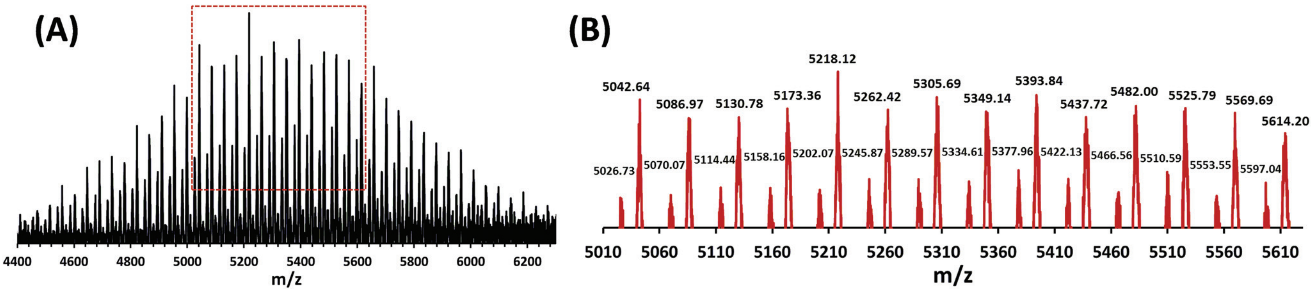


Fig. 7 MALDI-TOF mass spectrum for (A) mono-functional fraction collected at 8.45–8.70 min of MPEGCHO 5k under LCCC on XB-phenyl column at 30 °C, (B) enlarged part from 5010 to 5620 *m/z*.

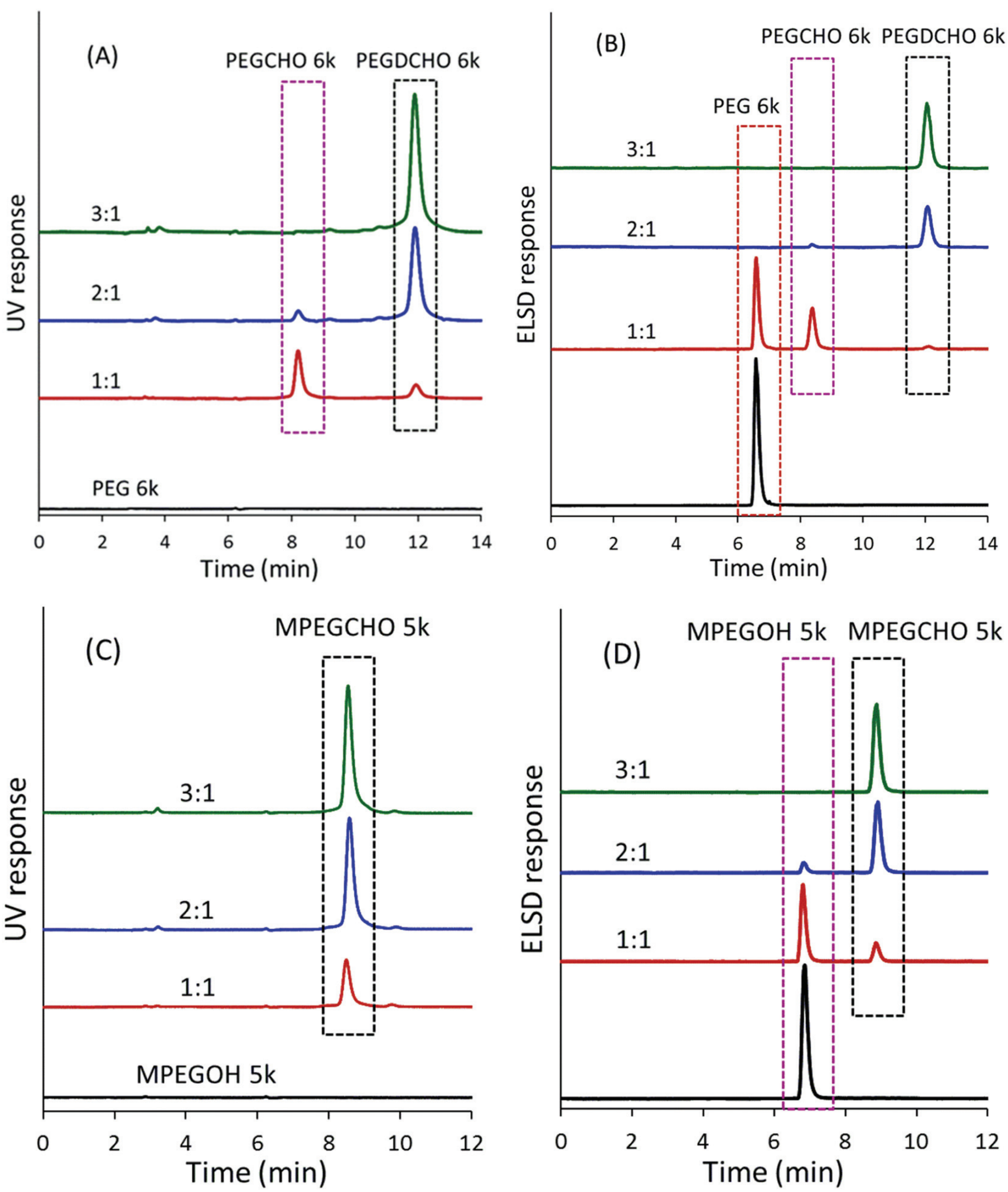


Fig. 8 The chromatograms of the obtained benzaldehyde-substituted PEG products at various feed ratios ([COOH]/[OH] indicated) on XB-phenyl column at 30 °C (mobile phase: acetonitrile : water = 45 : 55). (A, C) UV detection at 254 nm, (B, D) ELSD response, (A, B) starting polymer PEG 6k, (C, D) starting polymer MPEGOH 5k.

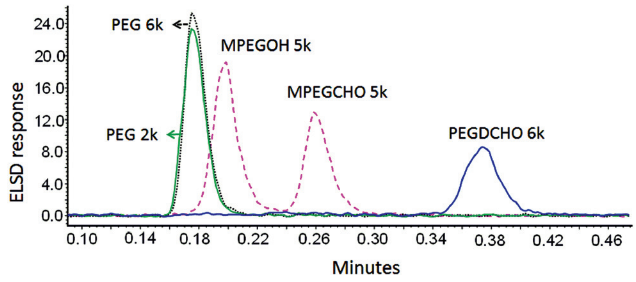


Fig. 9 The chromatograms of the PEGs and benzaldehyde-substituted PEGs by UPLC.

were observed with a repeating unit of 44 Da, corresponding to a single CH_2CH_2O monomeric unit. The major series results from the MPEGCHO 5k with one potassium cation (K^+). The minor series with 16 Da less (relative to the K^+ series) arises from the MPEGCHO 5k with one sodium cation (Na^+). The calculated mass of the end groups ($M_{\text{end group}}$ = 164) corresponds to the polymeric structure with COPhCHO as one end group and CH_3O as the other end group. Similar MALDI-TOF-MS spectra for bi-functional fraction PEGDCHO 6k and mono-functional fraction PEGCHO 6k are shown in Fig. S1 and S2† with both potassium cation main series and sodium cation minor series.

To optimize the synthesis conditions for benzaldehyde-substituted PEGs, especially the feed ratio of COOH (4-carboxybenzaldehyde) and OH (PEG), we investigated the non-, mono- and bi-functional distributions of the obtained benzaldehyde-substituted PEG products at various feed ratios by the LCCC. As illustrated in Fig. 8, it is clearly shown that the proportion of benzaldehyde-substituted PEGs varied along with the feed ratio of [COOH]/[OH] (1–3). When the feed ratio of [COOH]/[OH] was 1, the main part was unreacted PEGs with some mono-functional PEGCHO 6k and the minor part of bi-functional PEGDCHO 6k. When the feed ratio of [COOH]/[OH] was 2, the main part was bi-functional PEGDCHO 6k with the minor part of mono-functional PEGCHO 6k and negligible amounts of unreacted PEGs. When the feed ratio of [COOH]/[OH] was 3, the main product was bi-functional PEGDCHO 6k with negligible amounts of mono-functional PEGCHO 6k and unreacted PEGs. Similarly, as shown in Fig. 8C and D, MPEGOH 5k could be fully reacted to produce MPEGCHO 5k when the feed ratio of [COOH]/[OH] was at or above 3. Thus the LCCC could provide very useful information to optimize the synthesis of functional PEGs.

Separation of PEG derivatives by UPLC

Ultra-performance liquid chromatography (UPLC) is a relatively new technique giving new possibilities in liquid chromatography, especially concerning the decrease of analysis time and solvent consumption.[44–46] Separation of PEGs and benzaldehyde-substituted PEGs under LCCC was also studied by UPLC (with C18 column). As shown in Fig. 9, the separation was very quick and completed in less than half a minute, the PEGs and benzaldehyde-substituted PEGs were successfully separated according to the number of benzaldehyde end-groups by UPLC. The retention times were almost the same for PEG 2k and PEG 6k confirming no influence of the molar mass on the retention time. The bi-functional PEGDCHO 6k with stronger hydrophobic interactions showed a longer retention time than mono-functional MPEGCHO 5k. The separation mechanism is the same for HPLC and UPLC, however UPLC could significantly save the analysis time.

Conclusions

Benzaldehyde-substituted PEGs with various end-group functional degrees were synthesized. The baseline separation of PEGs and benzaldehyde-substituted PEG derivatives on reversed phase columns was achieved by normal HPLC and UPLC, according to the number of benzaldehyde end-groups without obvious effects of molar mass under the critical conditions. Off-line MALDI-TOF mass spectrometry was used to identify the repeating units and, especially, the end-groups of the benzaldehyde-substituted PEGs and further to confirm that this separation is based on the number of end functional groups of benzaldehyde. The separation was mainly due to the hydrophobic interaction between the benzaldehyde end-groups and the column packing. The retention times of benzaldehyde-substituted PEG derivatives increased with the increasing column temperature and ammonium acetate concentration, and with decreasing organic additive acetonitrile content in the aqueous mobile phase on reversed phase columns. LCCC could provide us very useful information to optimize the synthesis of functional PEGs. So far as we have known, this is the first report on separation of benzaldehyde-functional PEGs based on the number of benzaldehyde end-groups independent of the molar mass by isocratic elution using reversed-phase packing columns.

Acknowledgements

This research was financially supported by the National Natural Science Foundation of China (21374083, 21674083, 21174109, and 51533006).

Notes and references

1 P. L. Turecek, M. J. Bossard, F. Schoetens and I. A. Ivens, *J. Pharm. Sci.*, 2016, **105**, 460.
2 J. S. Suk, Q. G. Xu, N. Kim, J. Hanes and L. M. Ensign, *Adv. Drug Delivery Rev.*, 2016, **99**, 28.
3 H. Q. Dong, M. Tang, Y. Li, Y. Y. Li, D. Qian and D. L. Shi, *Nanomedicine*, 2015, **10**, 1941.
4 J. M. Harris, *Poly(ethylene glycol) chemistry: biotechnical and biomedical applications*, Springer, 2013.
5 X.-W. Lou, J. L. J. van Dongen and E. W. Meijer, *J. Chromatogr. A*, 2012, **1237**, 72.

6 A. A. D'souza and R. Shegokar, *Expert Opin. Drug Delivery*, 2016, **13**, 1.
7 A. Abuchowski, J. R. McCoy, N. C. Palczuk, T. van Es and F. F. Davis, *J. Biol. Chem.*, 1977, **252**, 3582.
8 T. Su, H. Yang, Q. Fan, D. L. Jia, Z. Tao, L. Wan and X. F. Lu, *Int. J. Pharm.*, 2016, **499**, 195.
9 M. Roberts, M. Bentley and J. Harris, *Adv. Drug Delivery Rev.*, 2012, **64**, 116.
10 E. Bakaic, N. M. B. Smeets and T. Hoare, *RSC Adv.*, 2015, **5**, 35469.
11 S. Lowe, N. M. O'Brien-Simpson and L. A. Connal, *Polym. Chem.*, 2015, **6**, 198.
12 D. D. McKinnon, D. W. Domaille, J. N. Cha and K. S. Anseth, *Adv. Mater.*, 2014, **26**, 865.
13 I. Yildiz, *J. Phys. Chem. A*, 2016, **120**, 3683.
14 D. Ossipov, S. Kootala, Z. Yi, X. Yang and J. Hilborn, *Macromolecules*, 2013, **46**, 4105.
15 K. Knop, R. Hoogenboom, D. Fischer and U. S. Schubert, *Angew. Chem., Int. Ed.*, 2010, **49**, 6288.
16 V. Rao N, S. Mane, A. Kishore, J. Das Sarma and R. Shunmugam, *Biomacromolecules*, 2011, **13**, 221.
17 F. Q. Hu, P. Meng, Y. Q. Dai, Y. Z. Du, J. You, X. H. Wei and H. Yuan, *Eur. J. Pharm. Biopharm.*, 2008, **70**, 749.
18 T. C. Tseng, L. Tao, F. Y. Hsieh, Y. Wei, I. M. Chiu and S. H. Hsu, *Adv. Mater.*, 2015, **27**, 3518.
19 Y. L. Zhang, L. Tao, S. X. Li and Y. Wei, *Biomacromolecules*, 2011, **12**, 2894.
20 T. Macko and D. Hunkeler, *Adv. Polym. Sci.*, 2003, **163**, 62.
21 H. Pasch and B. Trathnigg, *HPLC of Polymers*, Springer, 1999.
22 J. R. Montenegro-Burke, J. M. Bennett, J. A. McLean and D. M. Hercules, *Anal. Bioanal. Chem.*, 2016, **408**, 677.
23 A. A. Gorbunov and A. V. Vakhrushev, *J. Chromatogr. A*, 2015, **1456**, 162.
24 D. Mekap, T. Macko, R. Brull, R. Cong, A. W. deGroot, A. Parrott, P. Cools and W. Yau, *Polymer*, 2013, **54**, 5518.
25 C. Appiah, K. R. Siefermann, M. Jorewitz, H. Barqawi and W. H. Binder, *RSC Adv.*, 2016, **6**, 6358.
26 W. Radke, *J. Chromatogr. A*, 2014, **1335**, 62.
27 A. Baumgaertel, E. Altuntas and U. S. Schubert, *J. Chromatogr. A*, 2012, **1240**, 1.
28 X. Jiang, P. J. Schoenmakers, J. L. van Dongen, X. Lou, V. Lima and J. Brokken-Zijp, *Anal. Chem.*, 2003, **75**, 5517.
29 X. Jiang, P. J. Schoenmakers, X. Lou, V. Lima, J. L. van Dongen and J. Brokken-Zijp, *J. Chromatogr. A*, 2004, **1055**, 123.
30 X. Jiang, V. Lima and P. J. Schoenmakers, *J. Chromatogr. A*, 2003, **1018**, 19.
31 J. H. Arndt, T. Macko and R. Brüll, *J. Chromatogr. A*, 2013, **1310**, 1.
32 J. Gu, W.-P. Cheng, J. Liu, S.-Y. Lo, D. Smith, X. Qu and Z. Yang, *Biomacromolecules*, 2007, **9**, 255.
33 S. S. Bhati, T. Macko and R. Brull, *J. Chromatogr. A*, 2016, **1451**, 91.
34 E. Uliyanchenko, S. van der Wal and P. J. Schoenmakers, *Polym. Chem.*, 2012, **3**, 2313.
35 Y.-Z. Wei, R.-X. Zhuo and X.-L. Jiang, *J. Sep. Sci.*, 2016, DOI: 10.1002/jssc.201600904.
36 Y.-Z. Wei, R.-X. Zhuo and X.-L. Jiang, *J. Chromatogr. A*, 2016, **1447**, 122.
37 B. Trathnigg, N. V. Cuong and H. Ahmed, *J. Sep. Sci.*, 2009, **32**, 2857.
38 N. V. Cuong and B. Trathnigg, *J. Sep. Sci.*, 2010, **33**, 1052.
39 C. H. Lochmuller, M. A. Moebus, Q. C. Liu, C. Jiang and M. Elomaa, *J. Chromatogr. Sci.*, 1996, **34**, 69.
40 D. Haidacher, A. Vailaya and C. Horvath, *Proc. Natl. Acad. Sci. U. S. A.*, 1996, **93**, 2290.
41 B. N. Barman, D. H. Champion and S. L. Sjoberg, *J. Chromatogr. A*, 2009, **1216**, 6816.
42 M. Liu, C. Xie, H. Pan, J. Pan and W. Lu, *J. Chromatogr. A*, 2006, **1129**, 61.
43 M. Yang, S. Fazio, D. Munch and P. Drumm, *J. Chromatogr. A*, 2005, **1097**, 124.
44 M. E. Swartz, *J. Liq. Chromatogr. Relat. Technol.*, 2005, **28**, 1253.
45 E. Uliyanchenko, P. J. Schoenmakers and S. van der Wal, *J. Chromatogr. A*, 2011, **1218**, 1509.
46 M. J. Motilva, A. Serra and A. Macia, *J. Chromatogr. A*, 2013, **1292**, 66.

